

AI FAQ ASSISTANT API

- **Project Title:** "Ai Faq Assistant Api"
- **Team ID:** NM2025TMID38548
- **Team Leader:** M. Divya sree
- **Mail id:** sreed1347@gmail.com [Coder]
- **Team Members:**
 - NAME: M.Anusree
 - MAILID: anusreemuruganandham2007@gmail.com [Coder]
 - NAME: N.Deepika
 - MAIL ID: deepikapooja2728@gmail.com [Editing]
 - NAME: M.Dhivya sree
 - MAIL ID: dhivyaaleela30@gmail.com [Documenter]
 - NAME: A.Ambeeleswari
 - MAIL ID: ambeeleswari14@gmail.com [Documenter]

REQUIREMENT ANALYSIS PHASE

AI FAQ Assistant API

- ❖ The **AI FAQ Assistant API** is designed to provide users with quick and relevant answers to frequently asked questions. The system uses an AI-based approach to understand user queries and provide suitable responses from the available FAQ information. The API acts as a bridge between the frontend application, backend services, database, and AI processing system.

1. Objectives

The main objectives of the AI FAQ Assistant API are:

- To provide automated answers to frequently asked questions.
- To allow users to ask questions using natural language.
- To process user queries and identify relevant information.
- To store and retrieve FAQ data efficiently.
- To provide secure user authentication.
- To reduce the time and effort required to search for information.
- To provide APIs that can be easily integrated with web applications.

2. Functional Requirements

The system should provide the following functionalities:

- **User Registration and Login:** Users should be able to register using their basic details and securely log in to the application.
- **FAQ Management:** The system should allow authorized users to create, view, update, and delete FAQ information.
- **Question Processing:** Users should be able to submit questions through the API. The system should process the question and identify the relevant FAQ information.
- **AI Response:** The AI component should understand the user's query and provide a relevant and understandable response. If a suitable answer is not available, the system should return an appropriate message.
- **Database Management:** User details, FAQ questions, answers, and other required information should be stored and retrieved from the database.

3. Non-Functional Requirements

- ❖ The API should provide **good performance** and return responses within a reasonable time. It should maintain **security** by protecting user information and authentication details. The system should be **reliable** and handle invalid requests and errors properly. It should also be **scalable** so that more users and FAQ data can be supported in the future. The API structure should be simple and **maintainable** for future development.

4. Technical Requirements

- The proposed system uses **Node.js and Express.js** for backend development, **MongoDB** for database management, and **JavaScript** as the programming language. **React.js** can be used for the frontend application. **Postman** can be used for API testing, while **Git and GitHub** can be used for version control. AI/NLP services are used for processing user questions and generating relevant responses. Data is exchanged between the frontend and backend using **JSON** format.

5. API Requirements

- ✓ The API should provide separate endpoints for user authentication and FAQ operations.
- **User APIs:**
 - /api/user/register
 - /api/user/login
- **FAQ APIs:**
 - FAQ creation, retrieval, updating, and deletion endpoints, along with an endpoint for submitting questions to the AI assistant.
- The APIs should use appropriate HTTP methods such as **GET, POST, PUT, and DELETE** and should return proper status codes and JSON responses.

6. Conclusion

- The requirement analysis defines the essential features and technical needs of the **AI FAQ Assistant API**. The system should be secure, reliable, easy to use, and capable of providing relevant answers to user queries. These requirements provide a clear

foundation for the **system design, implementation, testing, and deployment** phases of the project.